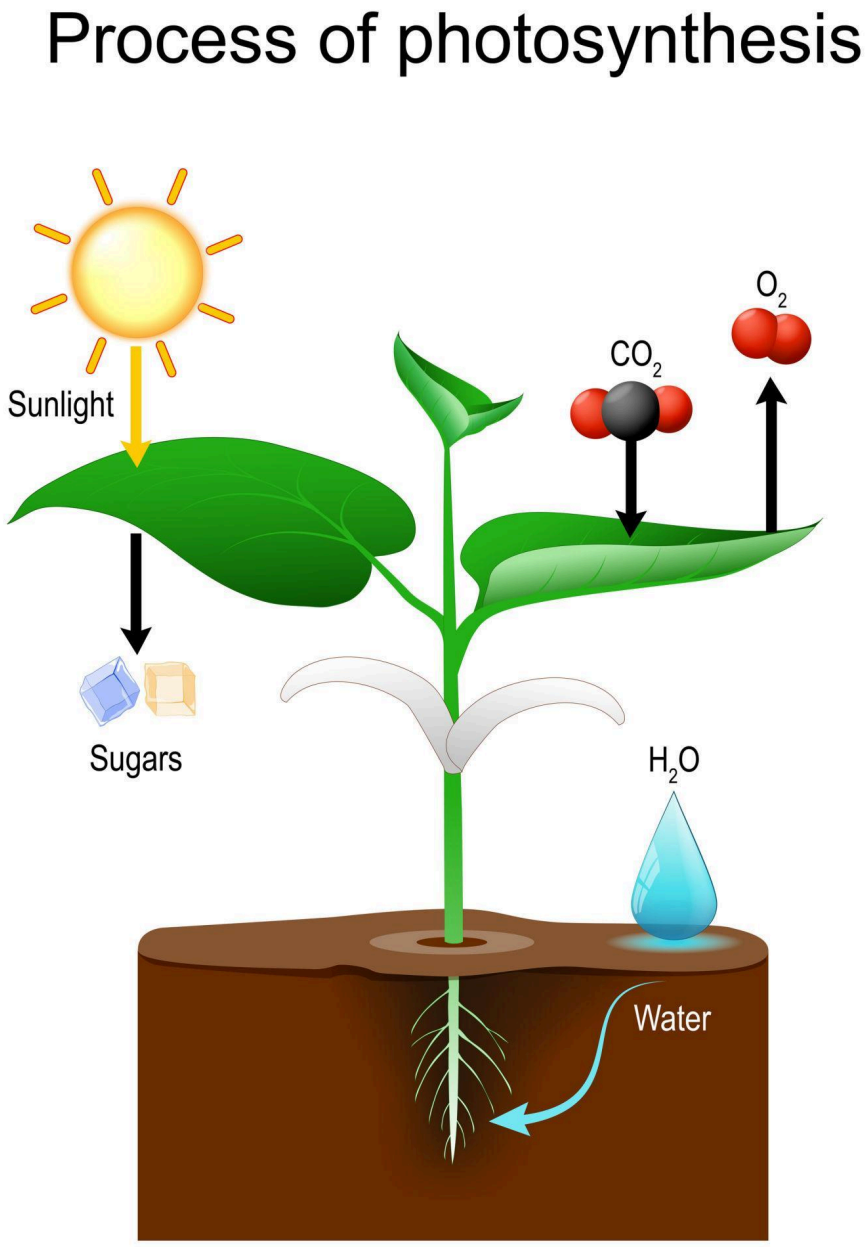
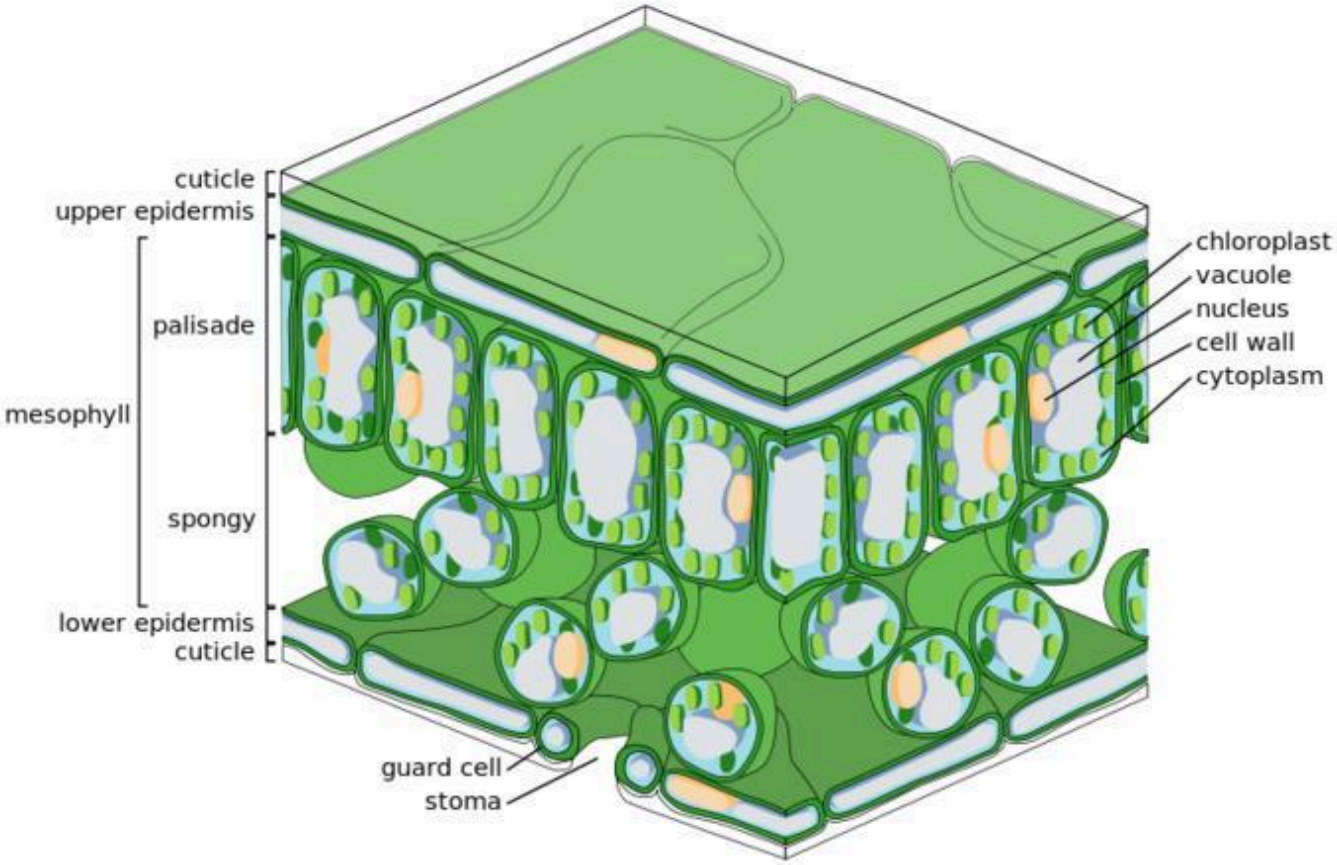
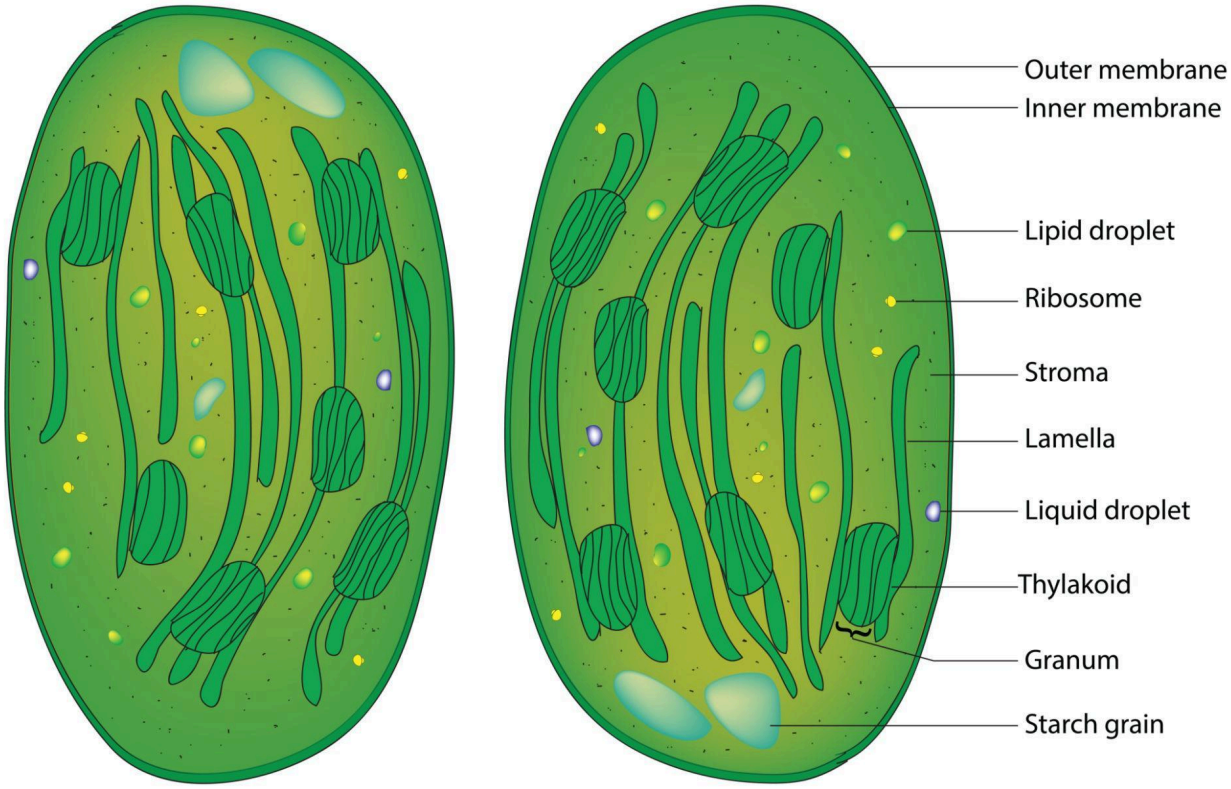
Photosynthesis — A Deep and Clear Explanation




cuticle
upper epidermis
palisade
mesophyll
spongy
lower epidermis
cuticle
guard cell
stoma
chloroplast
vacuole
nucleus
cell wall
cytoplasm

Structure of Chloroplast



Photosynthesis

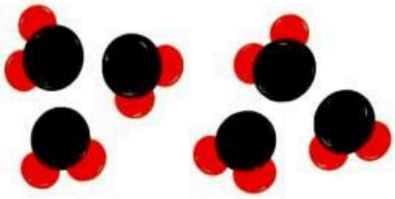
The steps

1 Light Absorption

Plant leaves contain a pigment called chlorophyll, which absorbs sunlight.

2 CO_2 Uptake

Carbon dioxide is taken in by plants through tiny openings on leaves called stomata.



3 Water Uptake

Plants absorb water from the soil through their roots and transport it to the leaves.

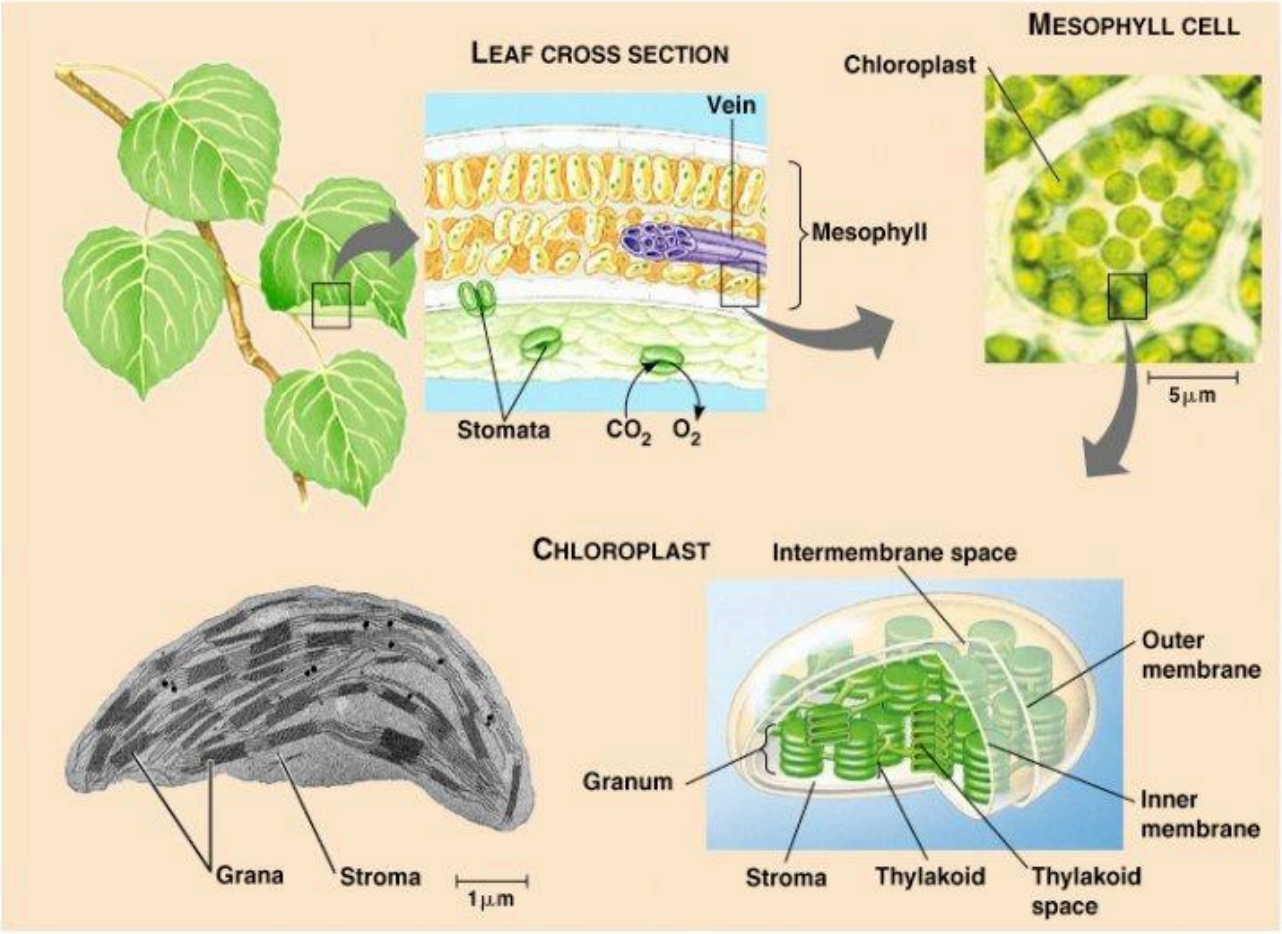
4 The Reaction

Photosynthesis occurs in chloroplasts, where sunlight energy is used to transform water and carbon dioxide into glucose and oxygen.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

5 Glucose Production and Oxygen Release

Photosynthesis produces glucose, which is the primary source of energy for plants, and releases oxygen as a byproduct



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Imagine a plant as a tiny food factory. Humans go to stores for food, animals eat plants or other animals, but plants have an extraordinary ability: **they create their own food from sunlight.**

This process is called **photosynthesis**.

The word itself can be broken down:

- **Photo** = light
- **Synthesis** = putting things together to make something

So photosynthesis means:

"Using light energy to create food."

Why do plants need photosynthesis?

Plants need energy just like humans do.

Humans obtain energy by eating:

- Rice
- Fruits

- Vegetables
- Bread

Plants cannot walk around and eat food. So they manufacture their own food by combining:

1. Sunlight ☀️
2. Water 💧
3. Carbon dioxide (CO_2) 🌫️

to produce:

1. Glucose (food) 🍬
2. Oxygen (O_2) 🌬️

The Overall Chemical Equation

This equation summarizes the entire process:

6CO_2+6H_2O\xrightarrow{light\ energy,chlorophyll}C_6H_{12}O_6+6O_2

Meaning:

- **6 CO_2** → six carbon dioxide molecules
- **6 H_2O** → six water molecules
- **Light energy + chlorophyll** help the reaction occur
- **$C_6H_{12}O_6$** → glucose (food)
- **6 O_2** → oxygen released

Where does photosynthesis happen?

Photosynthesis mainly occurs in the **leaves** of plants.

Leaves are called the **food factories of plants**.

Inside leaves are tiny structures called **chloroplasts**.

Inside chloroplasts is a green pigment called **chlorophyll**.

Role of chlorophyll:

- Absorbs sunlight
- Traps solar energy
- Converts light energy into chemical energy

Think of chlorophyll as a **solar panel in a plant**.

Structure involved in photosynthesis

1. Roots

Roots absorb:

- Water
- Minerals

Water travels upward through special tubes called **xylem**.

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2. Stomata

Leaves contain tiny pores called **stomata**.

Stomata function like tiny doors.

They:

- Allow CO_2 to enter
- Release O_2
- Control water loss

Guard cells around stomata open and close these pores.

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3. Chloroplast

The chloroplast is the plant cell organelle where photosynthesis occurs.

Inside chloroplasts:

- Chlorophyll captures sunlight
- Chemical reactions happen
- Glucose is formed

Step-by-step process of photosynthesis

Step 1: Absorption of sunlight

Chlorophyll absorbs light energy from the Sun.

Not all colors are absorbed equally.

- Green light is mostly reflected
- Red and blue light are absorbed efficiently

This is why plants appear green.

Step 2: Water absorption

Roots absorb water from soil.

Water moves through:

Root → Stem → Leaf

using xylem vessels.

Step 3: Carbon dioxide enters

Carbon dioxide from air enters through stomata.

Plants continuously exchange gases with the atmosphere.

Step 4: Light reaction

The first stage uses sunlight directly.

Major events:

A. Water splitting (Photolysis)

Water molecules split into:

- Hydrogen
- Oxygen
- Electrons

2H_2O\rightarrow4H^+ +4e^-+O_2

Results:

- Oxygen is released
- Electrons store energy
- Hydrogen helps make glucose

Step 5: Dark reaction (Calvin cycle)

The name "dark reaction" is slightly misleading.

It does **not require direct light**, but it usually occurs during daytime because it uses products generated in the light reaction.

During this stage:

- Carbon dioxide combines with hydrogen
- Glucose is produced

What happens to glucose?

Plants use glucose in several ways:

1. Energy production

Plants break glucose during respiration.

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2. Storage

Extra glucose becomes:

Starch

Examples:

- Potato stores starch
- Rice stores starch
- Wheat stores starch

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3. Growth

Glucose helps make:

- Cellulose
- Proteins
- Fats

Why oxygen is important

The oxygen released during photosynthesis is essential for life.

Animals and humans breathe oxygen:

Photosynthesis → Oxygen release → Respiration

Without photosynthesis:

- Oxygen levels would decrease
- Food chains would collapse
- Most life would not survive

Factors affecting photosynthesis

1. Light intensity

More light → faster photosynthesis (up to a limit)

Too much light may damage cells.

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2. Carbon dioxide concentration

More CO_2 generally increases photosynthesis until saturation.

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3. Temperature

Enzymes work best within an optimal temperature range.

Very high temperatures slow the process.

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4. Water availability

Less water:

- Stomata close
- CO_2 entry decreases
- Photosynthesis slows

Real-life examples

During daytime

Plants:

- Take in CO_2
- Release O_2

During night

Without sunlight:

- Photosynthesis stops
- Respiration continues

Plants then:

- Take in O_2
- Release CO_2

Why plants near homes feel fresh

Plants continuously contribute oxygen and improve air quality.

Importance of photosynthesis

Photosynthesis is important because it:

✓ Produces food for plants
✓ Supports the entire food chain
✓ Releases oxygen
✓ Maintains atmospheric balance
✓ Stores solar energy in food
✓ Reduces carbon dioxide

One-line summary

Photosynthesis is the process by which green plants use sunlight, water, and carbon dioxide to make glucose and release oxygen.

Everything we eat—directly or indirectly—begins with photosynthesis. It is one of the most important biological processes sustaining life on Earth.